

A Curvature-Deformed Arcsine Coordinate for Cubic Geometric Refinement

Exact Spherical Conjugacy, Riemannian Expansion Through Fifth Order,
and a High-Precision Weight-Six Reconstruction

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August 8, 2026

Abstract

A normalized geodesic side law on a Riemannian surface defines, by inversion, a local linearizing coordinate for geometric encode–scale–decode refinement. In the Euclidean limit this coordinate is the ordinary arcsine function. This paper develops the resulting *curvature-deformed arcsine*

$$\Psi_{\rho,w}(z) = \frac{1}{2} H_{\rho,w}^{-1}(2z),$$

where $H_{\rho,w}$ is the normalized signed side law for equal-radius geodesic endpoints based at p and centered about a unit tangent direction w .

The construction gives an exact Schröder/Koenigs conjugacy

$$\Psi_{\rho,w}(\mathcal{T}_{\rho,w,q}(z)) = q \Psi_{\rho,w}(z),$$

and hence an exact local multiplicative semigroup. On a sphere the deformed arcsine is available in closed form,

$$\Psi_{\alpha}(z) = \arcsin\left(\frac{\sin(z \sin \alpha)}{\sin \alpha}\right),$$

which interpolates between $\arcsin z$ in the flat limit and the identity map at the hemispherical value $\alpha = \pi/2$.

For a general Riemannian surface the variable-curvature expansion is derived analytically through geometric weight five. The coefficient functions are finite Laurent polynomials in $r = \sqrt{1 - z^2}$, and therefore every geometric-weight row of the Taylor coefficients is a finite combination of generalized binomial sequences. At weight six, where the available geodesic-distance expansion no longer directly supplies the required degree-eight side-law term, a high-precision intrinsic torus computation is used to recover a rational candidate formula in eight natural curvature invariants. Independent geometry and angle holdouts, including a failed restricted ansatz and its subsequent correction, provide a stringent computational validation. Conditional on that recovered side law, exact symbolic inversion yields the complete weight-six Laurent kernels and all Taylor coefficients at that weight.

The results motivate a finite Laurent-kernel conjecture for the full curvature hierarchy and extend the cubic N -series coefficient-lifting mechanism from the ordinary arcsine to its Riemannian deformation.

2020 Mathematics Subject Classification. 53B20, 53A55, 39B12, 65B05.

Key words and phrases. Riemannian normal coordinates, geodesic distance, Schröder equation, Koenigs linearization, Gaussian curvature, asymptotic expansion, geometric refinement.

1 Introduction

A recurring feature of cubic geometric refinement is that a nonlinear geometric construction becomes linear after passing to the correct local coordinate. In the flat circular problem that coordinate is the arcsine: the classical triple-angle polynomial

$$3z - 4z^3$$

is conjugate to multiplication by 3 through

$$\arcsin(3z - 4z^3) = 3 \arcsin z$$

on the appropriate local branch.

The same mechanism persists intrinsically on a Riemannian surface, but the linearizing coordinate is no longer the ordinary arcsine. The geometry itself deforms it.

Let p be a point of an oriented Riemannian surface, $w \in T_p M$ a unit tangent direction, and $\rho > 0$ a sufficiently small geodesic radius. The equal-radius signed side law

$$c_{\rho,w}(x)$$

records the signed intrinsic distance between radial geodesic endpoints whose angular separation is x , centered about the direction w . Normalize it by

$$H_{\rho,w}(x) = \frac{c_{\rho,w}(x)}{c'_{\rho,w}(0)}.$$

The associated geometric encode–scale–decode map at scale $q > 0$ is

$$\Phi_{\rho,w,q} = H_{\rho,w}^{-1} \circ (qH_{\rho,w}).$$

Thus

$$H_{\rho,w}(\Phi_{\rho,w,q}(x)) = qH_{\rho,w}(x)$$

holds identically wherever the local inverse is defined.

This observation has two consequences. First, the normalized side law is already a Koenigs coordinate for the refinement family, equivalently a local solution of Schröder’s equation; no separate linearization problem has to be solved. Second, after a half-angle normalization the flat coordinate becomes precisely $\arcsin z$. This leads to the definition

$$\Psi_{\rho,w}(z) = \frac{1}{2}H_{\rho,w}^{-1}(2z),$$

which is the central object of the present paper.

The work has four layers.

1. The normalized side law gives an exact local semigroup and an exact curvature-deformed arcsine coordinate.
2. On a sphere, the coordinate and the resulting scale maps are available in closed form.
3. On a general Riemannian surface, the variable-curvature deformation is expanded analytically through geometric weight five.

4. At weight six, an independent high-precision intrinsic torus calculation reconstructs a rational candidate for the new side-law term at high precision. Exact symbolic algebra then propagates that candidate to the full weight-six linearizer and to all Taylor degrees.

The fourth item is deliberately separated in status from the first three. The weight-six side-law coefficients have not yet been obtained from an independent degree-eight normal-coordinate derivation. They are therefore stated as a high-precision computational candidate. All algebraic consequences drawn from that candidate are exact.

The Riemann-normal-coordinate calculations through the lower orders use the corrected geodesic arc-length expansion developed by Brewin and its public source repository [1, 2]. The abstract local linearization mechanism belongs to the classical theory of the Schröder equation; see, for example, Iserles [3]. Likewise, the organization of normal-coordinate coefficients by natural curvature tensors is classical; a recent formulation is given by Graham and Kuo [4]. No novelty is claimed for those general frameworks. The specific subject here is their realization by this normalized intrinsic side law, the resulting spherical and variable-curvature linearizing coordinate, and the finite Laurent structure of its coefficient functions. The spherical, fifth-order curvature, and flat cubic coefficient-lifting results build on the author's preceding notes [5, 6, 7].

2 Geometric side law and exact linearization

All statements below are local. We take $\rho > 0$ sufficiently small that the relevant radial endpoints and their joining short geodesics remain in a common convex normal neighbourhood of p , and all inverse maps are taken on the local branch through the origin.

We fix the curvature convention

$$R(A, B)C = \nabla_A \nabla_B C - \nabla_B \nabla_A C - \nabla_{[A, B]} C, \quad \mathcal{R}(A, B, C, D) = \langle R(A, B)C, D \rangle,$$

so that on a surface

$$\mathcal{R}(A, B, B, A) = K(|A|^2|B|^2 - \langle A, B \rangle^2).$$

Thus, in components,

$$\mathcal{R}_{abcd} = K(g_{ad}g_{bc} - g_{ac}g_{bd}).$$

Brewin's displayed arc-length formulas use the opposite four-slot component sign in the slot convention employed in the lower-order derivation; when those formulas are invoked we therefore use

$$\tilde{R}_{abcd} := -\mathcal{R}_{abcd} = K(g_{ac}g_{bd} - g_{ad}g_{bc}).$$

This convention lock is the one used in the fifth-order source calculation [6].

Let (M, g) be a smooth oriented Riemannian surface. Fix

$$p \in M, \quad |w|_g = 1,$$

and let J denote rotation by $+\pi/2$ in the oriented tangent plane.

For sufficiently small x , define

$$u_x = \cos \frac{x}{2} w + \sin \frac{x}{2} Jw, \quad v_x = \cos \frac{x}{2} w - \sin \frac{x}{2} Jw.$$

Let

$$P_x = \exp_p(\rho u_x), \quad Q_x = \exp_p(\rho v_x).$$

The signed side law $c_{\rho,w}(x)$ is the signed length of the unique short geodesic from P_x to Q_x , with sign chosen so that

$$c_{\rho,w}(-x) = -c_{\rho,w}(x).$$

Hence $c_{\rho,w}$ is locally odd.

Define

$$H_{\rho,w}(x) = \frac{c_{\rho,w}(x)}{c'_{\rho,w}(0)}.$$

Then

$$H_{\rho,w}(0) = 0, \quad H'_{\rho,w}(0) = 1,$$

so $H_{\rho,w}$ has a local inverse. It is convenient to introduce the normalized half-side coordinate

$$S_{\rho,w}(u) := \frac{1}{2}H_{\rho,w}(2u).$$

Then $S_{\rho,w}(0) = 0$, $S'_{\rho,w}(0) = 1$, and the deformed arcsine introduced below is simply

$$\Psi_{\rho,w} = S_{\rho,w}^{-1}.$$

Proposition 2.1 (Exact geometric linearization). *For every admissible local scale $q > 0$, define*

$$\Phi_{\rho,w,q} = H_{\rho,w}^{-1} \circ (qH_{\rho,w}).$$

Then

$$H_{\rho,w}(\Phi_{\rho,w,q}(x)) = qH_{\rho,w}(x),$$

and the maps obey the local semigroup law

$$\Phi_{\rho,w,q} \circ \Phi_{\rho,w,s} = \Phi_{\rho,w,qs}.$$

Proof. Both statements follow directly from the definition and local invertibility of $H_{\rho,w}$. □

Writing $q = e^t$ gives a local flow

$$\Phi_t = H_{\rho,w}^{-1}(e^t H_{\rho,w}),$$

whose infinitesimal generator is

$$V_{\rho,w}(x) = \frac{H_{\rho,w}(x)}{H'_{\rho,w}(x)}.$$

In the flat circular case,

$$H_0(x) = 2 \sin \frac{x}{2}, \quad V_0(x) = 2 \tan \frac{x}{2}.$$

3 The curvature-deformed arcsine

Definition 3.1. The curvature-deformed arcsine associated with (p, w, ρ) is

$$\Psi_{\rho,w}(z) = \frac{1}{2}H_{\rho,w}^{-1}(2z).$$

Equivalently,

$$\Psi_{\rho,w} = S_{\rho,w}^{-1}, \quad \Psi_{\rho,w}^{-1}(y) = S_{\rho,w}(y) = \frac{1}{2}H_{\rho,w}(2y).$$

Define

$$\mathcal{T}_{\rho,w,q} = \Psi_{\rho,w}^{-1} \circ (q\Psi_{\rho,w}).$$

Theorem 3.2 (Exact Schröder relation). *For every admissible $q > 0$,*

$$\boxed{\Psi_{\rho,w}(\mathcal{T}_{\rho,w,q}(z)) = q\Psi_{\rho,w}(z).}$$

Moreover,

$$\mathcal{T}_{\rho,w,q} \circ \mathcal{T}_{\rho,w,s} = \mathcal{T}_{\rho,w,qs}.$$

In the flat limit,

$$H_0(x) = 2 \sin \frac{x}{2},$$

so

$$\boxed{\Psi_0(z) = \arcsin z.}$$

Thus the ordinary arcsine is the flat member of a geometrically defined family.

4 Exact spherical model

Let the surface be a sphere of radius R , and write

$$\alpha = \frac{\rho}{R}.$$

The normalized spherical side law is an elementary consequence of spherical trigonometry; see also the preceding spherical refinement note [5]. In the present setting its significance is that it gives the linearizing coordinate in closed form. Explicitly,

$$\boxed{H_\alpha(x) = \frac{2}{\sin \alpha} \arcsin\left(\sin \alpha \sin \frac{x}{2}\right).}$$

Theorem 4.1 (Exact spherical deformed arcsine). *For sufficiently small z ,*

$$\boxed{\Psi_\alpha(z) = \arcsin\left(\frac{\sin(z \sin \alpha)}{\sin \alpha}\right).}$$

The limiting cases are

$$\Psi_0(z) = \arcsin z, \quad \Psi_{\pi/2}(z) = z$$

on the principal local branch.

If $c = \cos^2 \alpha$, then

$$\Psi_\alpha(z) = z + \frac{c}{6}z^3 + \frac{c(c+8)}{120}z^5 + \frac{c(c^2+88c+136)}{5040}z^7 + O(z^9).$$

The exact spherical tripling map is

$$\mathcal{T}_\alpha = \Psi_\alpha^{-1} \circ (3\Psi_\alpha),$$

hence

$$\Psi_\alpha(\mathcal{T}_\alpha(z)) = 3\Psi_\alpha(z).$$

Its Taylor expansion begins

$$\mathcal{T}_\alpha(z) = 3z - 4cz^3 + 16c(c-1)z^5 - \frac{4}{15}c(c-1)(311c-221)z^7 + O(z^9).$$

At $c = 1$ this reduces to $3z - 4z^3$, while at $c = 0$ it reduces to $3z$.

5 General Riemannian expansion through weight five

Write

$$\begin{aligned} K &= K(p), & K_w &= (\nabla_w K)(p), & K_{ww} &= \nabla^2 K(w, w)(p), \\ D &= (\Delta K)(p), & D_w &= (\nabla_w \Delta K)(p), & T_{www} &= \nabla^3 K(w, w, w)(p). \end{aligned}$$

Let $r = \sqrt{1 - z^2}$. Then

$$\Psi_{\rho, w}(z) = \arcsin z + \rho^2 \Psi_2(z) + \rho^3 \Psi_3(z) + \rho^4 \Psi_4(z) + \rho^5 \Psi_5(z) + O(\rho^6).$$

5.1 Analytic side-law input and inversion

For completeness, we record the exact lower-order side-law input needed for the inversion. Put

$$s = \sin u, \quad c = \cos u.$$

The signed side law through fifth radial order, derived from the corrected normal-coordinate distance expansion in [1] with the convention lock above, is [6]

$$c_{\rho, w}(2u) = 2\rho s F_{\rho, w}(u),$$

where

$$F_{\rho, w}(u) = 1 + f_2 \rho^2 + f_3 \rho^3 + B_4 \rho^4 + B_5 \rho^5 + O(\rho^6),$$

with

$$\begin{aligned} f_2 &= -\frac{1}{6} K c^2, & f_3 &= -\frac{1}{12} K_w c^3, \\ B_4 &= \frac{c^2}{120} [(1 - 9s^2) K^2 - 3c^2 K_{ww} - s^2 (D - K_{ww})], \end{aligned}$$

and, writing

$$\begin{aligned} T_\Sigma &:= 3D_w - 3T_{www} + 2KK_w, \\ B_5 &= \frac{c^3}{1080} [(9c^2 - 92s^2) KK_w - 6c^2 T_{www} - 2s^2 T_\Sigma]. \end{aligned}$$

Since $c'_{\rho, w}(0) = \rho F_{\rho, w}(0)$, the normalized half-side coordinate is

$$S_{\rho, w}(u) = s \frac{F_{\rho, w}(u)}{F_{\rho, w}(0)}.$$

Thus the calculation of $\Psi_{\rho, w}$ through weight five is a pure perturbative inversion of this displayed scalar series.

Theorem 5.1 (Riemannian deformed arcsine through weight five). *The coefficient functions are*

$$\Psi_2(z) = -\frac{Kz^3}{6r} = -\frac{Kz}{6}(r^{-1} - r),$$

$$\Psi_3(z) = -\frac{K_w z(1 - r^3)}{12r} = -\frac{K_w z}{12}(r^{-1} - r^2),$$

$$\Psi_4(z) = \frac{z^3}{360r^3} [K^2(2r^4 + 13r^2 + 5) + 3Dr^4 - 3K_{ww}r^2(4r^2 + 3)],$$

and

$$\begin{aligned} \Psi_5(z) = \frac{z}{360r^3} & \left[2D_w r^5 z^2 \right. \\ & + K K_w (1 - r)(5r^6 + 5r^5 + 8r^4 + 18r^3 + 8r^2 + 5r + 5) \\ & \left. - 2T_{www} r^2 (1 - r)(2r^4 + 2r^3 + r^2 + r + 1) \right]. \end{aligned}$$

Proof. Write

$$u = \arcsin z + \rho^2 U_2 + \rho^3 U_3 + \rho^4 U_4 + \rho^5 U_5 + O(\rho^6)$$

and substitute into $S_{\rho,w}(u) = z$. At each radial order the unknown U_m occurs linearly with coefficient $\cos(\arcsin z) = r \neq 0$ on the local branch. Solving successively for U_2, U_3, U_4, U_5 and simplifying with $z^2 = 1 - r^2$ gives the four displayed coefficient functions. The calculation uses exact rational algebra; the accompanying symbolic audit reconstructs the normalized side law from the formulas above and verifies all four identities identically. $\square$

Equivalently,

$$\begin{aligned} \Psi_4(z) = \frac{K^2 z}{360} & (-2r^3 - 11r + 8r^{-1} + 5r^{-3}) \\ & + \frac{Dz}{120}(r - r^3) + \frac{K_{ww}z}{120}(4r^3 - r - 3r^{-1}), \end{aligned}$$

and

$$\begin{aligned} \Psi_5(z) = \frac{K K_w z}{360} & (-5r^4 - 3r^2 - 10r + 10 + 3r^{-1} + 5r^{-3}) \\ & + \frac{D_w z}{180}(r^2 - r^4) + \frac{T_{www} z}{360}(4r^4 - 2r^2 - 2r^{-1}). \end{aligned}$$

The D_w row terminates:

$$\frac{D_w z}{180}(r^2 - r^4) = \frac{D_w}{180}(z^3 - z^5),$$

so D_w contributes only to γ_3 and γ_5 .

5.2 First Taylor coefficients

Write

$$\Psi_{\rho,w}(z) = z + \gamma_3 z^3 + \gamma_5 z^5 + \gamma_7 z^7 + \cdots.$$

Through weight five,

$$\begin{aligned}
\gamma_3 &= \frac{1}{6} - \frac{1}{6}K\rho^2 - \frac{1}{8}K_w\rho^3 \\
&\quad + \left(\frac{1}{18}K^2 + \frac{1}{120}D - \frac{7}{120}K_{ww} \right) \rho^4 \\
&\quad + \left(\frac{3}{40}KK_w + \frac{1}{180}D_w - \frac{7}{360}T_{www} \right) \rho^5 + O(\rho^6), \\
\gamma_5 &= \frac{3}{40} - \frac{1}{12}K\rho^2 - \frac{1}{32}K_w\rho^3 \\
&\quad + \left(\frac{13}{360}K^2 - \frac{1}{240}D + \frac{1}{240}K_{ww} \right) \rho^4 \\
&\quad + \left(\frac{3}{160}KK_w - \frac{1}{180}D_w + \frac{13}{1440}T_{www} \right) \rho^5 + O(\rho^6),
\end{aligned}$$

and

$$\begin{aligned}
\gamma_7 &= \frac{5}{112} - \frac{1}{16}K\rho^2 - \frac{5}{192}K_w\rho^3 \\
&\quad + \left(\frac{7}{180}K^2 - \frac{1}{960}D - \frac{1}{192}K_{ww} \right) \rho^4 \\
&\quad + \left(\frac{5}{144}KK_w - \frac{1}{576}T_{www} \right) \rho^5 + O(\rho^6).
\end{aligned}$$

6 Curved tripling and cubic coefficient lifting

For the cubic scale set

$$\mathcal{T}_{\rho,w} = \mathcal{T}_{\rho,w,3} = \Psi_{\rho,w}^{-1} \circ (3\Psi_{\rho,w}).$$

Writing

$$\mathcal{T}_{\rho,w}(z) = 3z + \tau_3 z^3 + \tau_5 z^5 + \tau_7 z^7 + \cdots,$$

formal conjugacy gives

$$\boxed{\tau_3 = -24\gamma_3},$$

$$\boxed{\tau_5 = 24(27\gamma_3^2 - 10\gamma_5)},$$

and

$$\boxed{\tau_7 = -24(945\gamma_3^3 - 675\gamma_3\gamma_5 + 91\gamma_7)}.$$

The flat specialization gives $\mathcal{T}_0(z) = 3z - 4z^3$ exactly.

For $j \geq 1$, define

$$\mathcal{L}_j[P](z) = \frac{3^{2j+1}P(z) - P(\mathcal{T}_{\rho,w}(z))}{3(3^{2j} - 1)}.$$

Theorem 6.1 (Curved cubic coefficient lifting). *Let P_j be odd and suppose*

$$\Psi_{\rho,w}(z) - P_j(z) = O(z^{2j+1}).$$

Then

$$P_{j+1} = \mathcal{L}_j[P_j]$$

satisfies

$$\Psi_{\rho,w}(z) - P_{j+1}(z) = O(z^{2j+3}).$$

Proof. Set $E_j = \Psi_{\rho,w} - P_j$. By the exact Schröder relation,

$$\Psi_{\rho,w}(\mathcal{T}_{\rho,w}(z)) = 3\Psi_{\rho,w}(z).$$

If

$$E_j(z) = az^{2j+1} + O(z^{2j+3}),$$

then, because $\mathcal{T}_{\rho,w}(z) = 3z + O(z^3)$,

$$E_j(\mathcal{T}_{\rho,w}(z)) = a(3z)^{2j+1} + O(z^{2j+3}).$$

The defining linear combination cancels the leading error, and oddness skips the intervening even degree. $\square$

Thus the flat cubic N -series reconstructs the ordinary arcsine coefficient sequence, while the Riemannian cubic N -series reconstructs the curvature-deformed sequence.

7 Weight-six extension

Define

$$S_4 := \text{Sym } \nabla^4 K,$$

with contractions

$$S_{www} := S_4(w, w, w, w), \quad (\text{tr } S)_{ww} := (\text{tr } S_4)(w, w), \quad \text{tr}^2 S := \text{tr}(\text{tr } S_4).$$

The high-precision recovery is organized in the following natural weight-six family:

$$K^3, \quad K\Delta K, \quad KK_{ww}, \quad |\nabla K|^2, \quad K_w^2, \\ S_{www}, \quad (\text{tr } S)_{ww}, \quad \text{tr}^2 S.$$

This family is motivated by geometric weight, direction/orientation symmetry, and the classical natural-tensor framework [4]. At this stage it is a candidate basis for this side law rather than an abstract completeness theorem for all possible weight-six natural scalars. The full-rank recovery design and unused holdouts test the basis empirically.

7.1 High-precision side-law reconstruction

Let $z = \sin u$ and $r = \cos u$. Write

$$S_{\rho,w}(u) = \sin u + \rho^2 s_2(u) + \cdots + \rho^6 s_6(u) + O(\rho^7).$$

Proposition 7.1 (High-precision weight-six reconstruction). *High-precision intrinsic torus computations, rational reconstruction, and independent geometry and angle holdout tests identify*

$$s_6 = z \left\{ K^3(1-r^2) \left(\frac{31}{15120} - \frac{11}{280}r^2 + \frac{5}{112}r^4 \right) \right. \\
+ K\Delta K(1-r^2) \left(-\frac{1}{112}r^2 + \frac{127}{15120}r^4 \right) \\
+ KK_{ww}(1-r^2) \left(\frac{29}{5040} + \frac{53}{5040}r^2 - \frac{53}{1260}r^4 \right) \\
+ |\nabla K|^2(1-r^2) \left(-\frac{17}{10080}r^2 + \frac{71}{30240}r^4 \right) \\
+ K_w^2 \left(\frac{5}{1008} + \frac{17}{10080}r^2 - \frac{1}{144}r^3 - \frac{157}{5040}r^4 + \frac{317}{10080}r^6 \right) \\
+ S_{wwww}(1-r^2) \left(\frac{1}{1008} + \frac{1}{1260}r^2 + \frac{1}{315}r^4 \right) \\
+ (\text{tr } S)_{ww}(1-r^2) \left(\frac{1}{2520}r^2 - \frac{1}{420}r^4 \right) \\
\left. + \text{tr}^2 S(1-r^2) \left(-\frac{1}{5040}r^2 + \frac{1}{5040}r^4 \right) \right\}.$$

Remark 7.2 (Status). Proposition 7.1 is a high-precision rational reconstruction supported by independent holdout computations; it is not yet an analytic degree-eight normal-coordinate theorem or an interval-arithmetic proof. The distinction is maintained throughout this paper.

The K^3 row is independently fixed by the exact spherical model:

$$[s_6]_{K^3} = \frac{K^3 z^3}{15120} (675r^4 - 594r^2 + 31).$$

The K_w^2 row deserves separate attention. A first restricted ansatz in which every row factored as

$$z(1-r^2)(A + Br^2 + Cr^4)$$

failed at new side angles. The discrepancy was traced to normalization of the already-known weight-three side-law term

$$f_3(u) = -\frac{K_w}{12}r^3.$$

Indeed,

$$f_3(0)^2 - f_3(u)f_3(0) = \frac{K_w^2}{144}(1-r^3),$$

which is not divisible by $1-r^2$ with an even-polynomial quotient. Once this forced nonlinear contribution is included, the independent angle holdouts return to the numerical extraction scale.

7.2 Universal inverse formula at weight six

Let

$$S_{\rho,w}(u) = s_0(u) + \rho^2 s_2(u) + \cdots + \rho^6 s_6(u) + O(\rho^7), \quad s_0(u) = \sin u,$$

and

$$\Psi_{\rho,w}(z) = u_0 + \rho^2 U_2 + \cdots + \rho^6 U_6 + O(\rho^7), \quad u_0 = \arcsin z.$$

Exact formal inversion gives

$$U_6 = -\frac{1}{r} \left[s_6 + s'_4 U_2 + s'_3 U_3 + s'_2 U_4 + \frac{1}{2} s''_2 U_2^2 - z \left(U_2 U_4 + \frac{1}{2} U_3^2 \right) - \frac{r}{6} U_2^3 \right],$$

where side-law derivatives are evaluated at $u = u_0$. In particular,

$$\boxed{s_5 \text{ and } U_5 \text{ do not enter } U_6.}$$

Thus

$$\Psi_6 = \Psi_6^{\text{ind}} - \frac{s_6}{r},$$

where the induced term is determined entirely by weights 2, 3, 4.

7.3 Complete finite Laurent layer

Theorem 7.3 (Conditional weight-six Laurent expansion). *Assume Proposition 7.1. Then*

$$\boxed{\Psi_6(z; p, w) = z \sum_I \mathcal{I}_I(p, w) L_I(r), \quad r = \sqrt{1 - z^2},}$$

with

$$\begin{aligned} L_{K^3} &= \frac{r^5}{2835} + \frac{r^3}{5670} + \frac{23r}{15120} + \frac{1}{9072r} + \frac{1}{6480r^3} - \frac{1}{432r^5}, \\ L_{K\Delta K} &= -\frac{r^5}{756} + \frac{53r^3}{15120} - \frac{r}{280} + \frac{1}{720r}, \\ L_{KK_{ww}} &= -\frac{r^5}{315} - \frac{5r^3}{504} + \frac{13r}{1680} + \frac{1}{840r} + \frac{1}{240r^3}, \\ L_{|\nabla K|^2} &= \frac{71r^5}{30240} - \frac{61r^3}{15120} + \frac{17r}{10080}, \\ L_{K_w^2} &= -\frac{r^5}{140} + \frac{139r^3}{10080} - \frac{r^2}{48} - \frac{17r}{10080} \\ &\quad + \frac{1}{72} - \frac{1}{672r} + \frac{1}{288r^3}, \\ L_{S_{www}} &= \frac{r^5}{315} - \frac{r^3}{420} + \frac{r}{5040} - \frac{1}{1008r}, \\ L_{(\text{tr } S)_{ww}} &= -\frac{r^5}{420} + \frac{r^3}{360} - \frac{r}{2520}, \end{aligned}$$

and

$$L_{\text{tr}^2 S} = \frac{r^5}{5040} - \frac{r^3}{2520} + \frac{r}{5040}.$$

Each kernel obeys $L_I(1) = 0$.

Proof. Using Proposition 7.1, substitute the previously established U_2, U_3, U_4 into the universal weight-six inverse formula. Exact symbolic simplification gives the displayed kernels. This inversion step uses no numerical approximation. $\square$

7.4 Exact spherical checksum at weight six

The constant-curvature sector of Theorem 7.3 can be checked independently from the exact spherical formula, without using the torus recovery. Expanding

$$\Psi_\alpha(z) = \arcsin\left(\frac{\sin(z \sin \alpha)}{\sin \alpha}\right)$$

through order α^6 gives

$$\boxed{[\alpha^6]\Psi_\alpha(z) = z \left(\frac{r^5}{2835} + \frac{r^3}{5670} + \frac{23r}{15120} + \frac{1}{9072r} + \frac{1}{6480r^3} - \frac{1}{432r^5} \right)}, \quad r = \sqrt{1 - z^2}.$$

Thus

$$[\alpha^6]\Psi_\alpha(z) = zL_{K^3}(r)$$

identically as a function of z . Since on a constant-curvature surface

$$\alpha = \rho\sqrt{K}, \quad \alpha^6 = K^3\rho^6,$$

the entire $K^3\rho^6$ row in the general weight-six Laurent expansion is fixed independently by the exact spherical model. Exact symbolic simplification of the difference between the two expressions gives zero.

8 All Taylor degrees at weight six

Define

$$\boxed{\beta_n^{(\nu)} = (-1)^n \binom{\nu/2}{n}},$$

so that

$$(1 - z^2)^{\nu/2} = \sum_{n \geq 0} \beta_n^{(\nu)} z^{2n}.$$

Corollary 8.1 (All- n weight-six coefficient rows). *Assume Proposition 7.1. For every $n \geq 1$,*

$$\begin{aligned} [\rho^6]\gamma_{2n+1} &= A_n K^3 + B_n K \Delta K + C_n K K_{ww} + D_n |\nabla K|^2 \\ &\quad + E_n K_w^2 + F_n S_{www} + G_n (\text{tr } S)_{ww} + H_n \text{tr}^2 S, \end{aligned}$$

where

$$\begin{aligned} A_n &= \frac{1}{2835} \beta_n^{(5)} + \frac{1}{5670} \beta_n^{(3)} + \frac{23}{15120} \beta_n^{(1)} \\ &\quad + \frac{1}{9072} \beta_n^{(-1)} + \frac{1}{6480} \beta_n^{(-3)} - \frac{1}{432} \beta_n^{(-5)}, \\ B_n &= -\frac{1}{756} \beta_n^{(5)} + \frac{53}{15120} \beta_n^{(3)} - \frac{1}{280} \beta_n^{(1)} + \frac{1}{720} \beta_n^{(-1)}, \\ C_n &= -\frac{1}{315} \beta_n^{(5)} - \frac{5}{504} \beta_n^{(3)} + \frac{13}{1680} \beta_n^{(1)} \\ &\quad + \frac{1}{840} \beta_n^{(-1)} + \frac{1}{240} \beta_n^{(-3)}, \\ D_n &= \frac{71}{30240} \beta_n^{(5)} - \frac{61}{15120} \beta_n^{(3)} + \frac{17}{10080} \beta_n^{(1)}, \end{aligned}$$

$$\begin{aligned}
E_n &= -\frac{1}{140}\beta_n^{(5)} + \frac{139}{10080}\beta_n^{(3)} - \frac{1}{48}\beta_n^{(2)} - \frac{17}{10080}\beta_n^{(1)} \\
&\quad + \frac{1}{72}\beta_n^{(0)} - \frac{1}{672}\beta_n^{(-1)} + \frac{1}{288}\beta_n^{(-3)}, \\
F_n &= \frac{1}{315}\beta_n^{(5)} - \frac{1}{420}\beta_n^{(3)} + \frac{1}{5040}\beta_n^{(1)} - \frac{1}{1008}\beta_n^{(-1)}, \\
G_n &= -\frac{1}{420}\beta_n^{(5)} + \frac{1}{360}\beta_n^{(3)} - \frac{1}{2520}\beta_n^{(1)},
\end{aligned}$$

and

$$H_n = \frac{1}{5040}\beta_n^{(5)} - \frac{1}{2520}\beta_n^{(3)} + \frac{1}{5040}\beta_n^{(1)}.$$

Proof. Every kernel in Theorem 7.3 is a finite linear combination of powers $r^\nu = (1 - z^2)^{\nu/2}$. Expanding each power by the generalized binomial theorem and collecting z^{2n+1} gives the result. $\square$

8.1 Low Taylor degrees through weight six

Combining the analytic lower-order results with the weight-six extension gives

$$\begin{aligned}
\gamma_3 &= \frac{1}{6} - \frac{1}{6}K\rho^2 - \frac{1}{8}K_w\rho^3 \\
&\quad + \left(\frac{1}{18}K^2 + \frac{1}{120}D - \frac{7}{120}K_{ww} \right) \rho^4 \\
&\quad + \left(\frac{3}{40}KK_w + \frac{1}{180}D_w - \frac{7}{360}T_{www} \right) \rho^5 \\
&\quad + \left(-\frac{1}{135}K^3 + \frac{1}{1890}K\Delta K + \frac{13}{504}KK_{ww} - \frac{1}{1512}|\nabla K|^2 \right. \\
&\quad \left. + \frac{47}{2016}K_w^2 - \frac{5}{1008}S_{www} + \frac{1}{504}(\text{tr } S)_{ww} \right) \rho^6 + O(\rho^7), \\
\gamma_5 &= \frac{3}{40} - \frac{1}{12}K\rho^2 - \frac{1}{32}K_w\rho^3 \\
&\quad + \left(\frac{13}{360}K^2 - \frac{1}{240}D + \frac{1}{240}K_{ww} \right) \rho^4 \\
&\quad + \left(\frac{3}{160}KK_w - \frac{1}{180}D_w + \frac{13}{1440}T_{www} \right) \rho^5 \\
&\quad + \left(-\frac{1}{108}K^3 - \frac{1}{5040}K\Delta K - \frac{1}{420}KK_{ww} + \frac{3}{1120}|\nabla K|^2 \right. \\
&\quad \left. - \frac{83}{40320}K_w^2 + \frac{47}{10080}S_{www} - \frac{17}{5040}(\text{tr } S)_{ww} + \frac{1}{5040}\text{tr}^2 S \right) \rho^6 + O(\rho^7),
\end{aligned}$$

and

$$\begin{aligned}
\gamma_7 = & \frac{5}{112} - \frac{1}{16}K\rho^2 - \frac{5}{192}K_w\rho^3 \\
& + \left(\frac{7}{180}K^2 - \frac{1}{960}D - \frac{1}{192}K_{ww} \right) \rho^4 \\
& + \left(\frac{5}{144}KK_w - \frac{1}{576}T_{www} \right) \rho^5 \\
& + \left(-\frac{227}{15120}K^3 + \frac{13}{10080}K\Delta K + \frac{3}{320}KK_{ww} - \frac{11}{10080}|\nabla K|^2 \right. \\
& \quad \left. + \frac{119}{11520}K_w^2 - \frac{59}{40320}S_{www} + \frac{19}{20160}(\text{tr } S)_{ww} - \frac{1}{10080}\text{tr}^2 S \right) \rho^6 + O(\rho^7).
\end{aligned}$$

At weight five,

$$\frac{D_w z}{180}(r^2 - r^4) = \frac{D_w}{180}(z^3 - z^5),$$

so $[D_w]\gamma_{2n+1} = 0$ for $n \geq 3$.

At weight six, $\beta_n^{(0)} = 0$ for $n \geq 1$, so the constant term in $L_{K_w^2}$ affects only the linear normalization. Also $\beta_n^{(2)} = 0$ for $n \geq 2$, so the term $-r^2/48$ contributes only to γ_3 . A further cancellation is

$$[\text{tr}^2 S \rho^6]\gamma_3 = 0.$$

9 Computational validation of the weight-six layer

The weight-six side-law candidate was recovered independently of the formal normal-coordinate inversion formulas using the ring-torus metric

$$ds^2 = a^2 d\theta^2 + (R + a \cos \theta)^2 d\phi^2.$$

Intrinsic radial endpoints and joining geodesics were computed directly at arbitrary precision.

The normalization derivative was obtained from the scalar Jacobi equation along the central radial geodesic,

$$j'' + Kj = 0, \quad j(0) = 0, \quad j'(0) = 1,$$

so that

$$c'_{\rho,w}(0) = j(\rho).$$

Opposite directions w and $-w$ were averaged, cancelling odd geometric weights. After subtraction of the known weight-two and weight-four terms,

$$E_6(\rho) = \frac{S_{\text{even}} - \sin u - \rho^2 s_2 - \rho^4 s_4}{\rho^6} = s_6 + O(\rho^2).$$

A higher even extrapolation basis

$$1, \rho^2, \rho^4, \rho^6, \rho^8$$

was used to recover s_6 . Ten torus geometries at three initial side angles produced thirty measurements. Twenty-one equations from the first seven geometries determined the unknown coefficients, while

nine equations from three unused geometries served as holdouts. After higher-order reprocessing, the holdout root-mean-square residual was approximately

$$9.6 \times 10^{-23}.$$

A second validation used side angles not present in the recovery. The initial restricted angular ansatz failed by many orders of magnitude beyond the extraction error. The discrepancy was proportional to K_w^2 across all tested geometries and was traced to the normalization term

$$\frac{K_w^2}{144}(1 - r^3).$$

After that mechanism was incorporated, twelve independent angle holdouts collapsed to a root-mean-square residual of approximately

$$3.5 \times 10^{-24}.$$

Finally, exact symbolic arithmetic verified the normalization

$$L_I(1) = 0,$$

the exact spherical K^3 kernel, and the all- n formulas against direct Taylor expansion through $n = 12$.

The accompanying reproducibility archive contains separate scripts for the weight-six inverse scaffold, angular-target and rank audits, high-precision torus pilot, full recovery, higher-order reprocessing, out-of-sample angular validation, direct K_w^2 kernel isolation, corrected-kernel check, and exact Laurent/all- n audit, together with the raw and processed data used for the reported holdouts.

10 Finite Laurent-kernel structure

Through weights two to six the deformed arcsine has the observed structure

$$\boxed{\Psi_m(z; p, w) = z \sum_I \mathcal{I}_{m,I}(p, w) L_{m,I}(\sqrt{1 - z^2}),}$$

where $\mathcal{I}_{m,I}$ is a natural curvature invariant of geometric weight m , and every coefficient function encountered so far is a finite Laurent polynomial.

The weight-six calculation is a stronger test than the lower orders because a more restrictive angular hypothesis was actually falsified. The failure was localized to a single invariant sector, explained by a nonlinear normalization effect already implied by the lower-order geometry, and corrected without changing the invariant family. Independent holdouts then returned to the numerical extraction scale.

Conjecture 10.1 (Finite Laurent-kernel conjecture). *At every finite geometric weight m ,*

$$\boxed{\Psi_m(z; p, w) = z \sum_I \mathcal{I}_{m,I}(p, w) L_{m,I}(r), \quad r = \sqrt{1 - z^2},}$$

where each $\mathcal{I}_{m,I}$ is an admissible natural curvature invariant of weight m , and each $L_{m,I}$ is a finite Laurent polynomial in r .

If the conjecture holds, then every geometric-weight row of

$$\gamma_3, \gamma_5, \gamma_7, \dots$$

is a finite linear combination of generalized binomial sequences

$$\beta_n^{(\nu)} = (-1)^n \binom{\nu/2}{n}.$$

Finite-support selection rules arise whenever a nonnegative integral exponent ν causes the associated generalized binomial sequence to terminate.

11 Discussion

The central point of this paper is not that local analytic linearization is new. Schröder and Koenigs linearization belong to classical local dynamics. What is geometrically significant here is that the normalized intrinsic side law supplies the linearizing coordinate directly:

$$H_{\rho,w}(\Phi_{\rho,w,q}(x)) = qH_{\rho,w}(x).$$

The corresponding half-angle inverse is a curvature-deformed arcsine.

The exact sphere provides a closed model in which curvature continuously deforms the flat arcsine toward the identity map. The general Riemannian expansion shows how the local curvature jet enters this coordinate. Through weight five the result is analytic. At weight six the intrinsic high-precision calculation reconstructs a candidate invariant layer whose subsequent Laurent and all- n consequences are exact conditional on that input.

The strongest structural observation is the persistence of finite Laurent kernels. This turns a local geometric expansion into a statement about an entire infinite family of Taylor coefficients at once. It also explains why selection rules can occur: finite nonnegative powers of r correspond to terminating generalized-binomial sequences.

The remaining major proof obligation is the independent derivation of the degree-eight geodesic-distance term needed to establish Proposition 7.1 analytically. Such a derivation would convert the present high-precision weight-six reconstruction into a theorem and would provide the first direct analytic test of Conjecture 10.1 beyond the orders from which the pattern was discovered.

12 Conclusion

The normalized geodesic side law defines an exact local linearizing coordinate for geometric refinement on a Riemannian surface. After half-angle normalization this coordinate becomes

$$\Psi_{\rho,w}(z) = \frac{1}{2} H_{\rho,w}^{-1}(2z),$$

which reduces to the ordinary arcsine in the flat limit.

The resulting framework gives:

- an exact local multiplicative semigroup;
- an exact spherical curvature-deformed arcsine;
- an analytic variable-curvature expansion through weight five;

- a curved cubic coefficient-lifting theorem;
- a high-precision reconstruction of the weight-six side-law candidate;
- exact weight-six finite Laurent kernels conditional on that candidate;
- all Taylor coefficients at weight six in generalized-binomial form; and
- a sharpened finite Laurent-kernel conjecture for the full curvature hierarchy.

The exact spherical solution, variable-curvature expansion, coefficient-lifting mechanism, and weight-six computational validation all point to a common interpretation: cubic geometric refinement does not lose the flat arcsine linearizer under curvature. Rather, curvature deforms that linearizer in a structured invariant hierarchy.

References

- [1] L. C. Brewin, “Riemann normal coordinate expansions using Cadabra,” *Classical and Quantum Gravity* **26** (2009), 175017; corrected arXiv version arXiv:0903.2087v4 (2022). doi:10.1088/0264-9381/26/17/175017.
- [2] L. C. Brewin, *riemann-normal-coords: LaTeX and Cadabra sources for Riemann normal coordinate expansions*, public GitHub repository, accessed August 8, 2026. <https://github.com/leo-brewin/riemann-normal-coords>.
- [3] A. Iserles, “A constructive approach to the Schröder equation,” *Journal of Computational and Applied Mathematics* **46** (1993), 301–314. doi:10.1016/0377-0427(93)90304-T.
- [4] C. R. Graham and T.-M. Kuo, “Geodesic normal coordinates and natural tensors for pseudo-Riemannian submanifolds,” *Proceedings of the American Mathematical Society* **154** (2026), 339–351. doi:10.1090/proc/17432.
- [5] W. Baker, “A Spherical Extension of the Baker Cubic Trisection Refinement Map,” unpublished manuscript, August 2026.
- [6] W. Baker, “Directional Curvature Expansion of the Baker Cubic Coefficient on Riemannian Surfaces Through Fifth Order,” unpublished manuscript, August 2026.
- [7] W. Baker, “Cubic N -Series Acceleration and Newton-Coefficient Lifting for the Arcsine,” unpublished manuscript, 2026.